Skills Summary

Graphing Basic Polar Curves from Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluate a polar equation at an angle

Idea: A polar equation $r = f(\theta)$ is a rule: feed it an angle, get a signed distance from the pole.

- The point is (r, θ): go out r along the ray at angle θ.
- **Negative r is legal.** Plot $|r|$ in the direction *opposite* θ; that is exactly how inner loops are drawn.
- Exact values only: $\cos 60° = \frac{1}{2}$, $\cos 120° = -\frac{1}{2}$, $\cos 180° = -1$.

Example: For $r = 3\cos\theta$, find r at $\theta = 60°$.
Step 1. A single angle is given and a distance is wanted, so this is plain substitution — no equation to solve.
Step 2. $r = 3\cos 60° = 3 \cdot \frac{1}{2}$. $\Rightarrow$ $\boldsymbol{\frac{3}{2}}$

Try it: For $r = 4\sin\theta$, find r at $\theta = 30°$.

check: **2**

Skill 2 — Pole crossings and maximum radius

Two questions that build the picture:

- **Zeros.** Solve $f(\theta) = 0$ on $0° \le \theta < 360°$. Each solution is an angle at which the curve passes through the pole — the boundary between petals, a cardioid's cusp, or the ends of an inner loop.
- **Maximum r.** Solve $f(\theta) = r_{\text{max}}$. Those angles point at the petal tips, and for a circle or limacon they give the direction the curve is offset from the pole.
- With $\sin(k\theta)$ or $\cos(k\theta)$, solve for $k\theta$ first, list every value up to $k \cdot 360°$, and only then divide by k — dividing too early loses solutions.

Example: Find every θ in $[0°, 360°)$ with $3\sin(2\theta) = 0$.
Step 1. Asking where the rose returns to the pole means solving $r = 0$; the doubled angle means the window for 2θ is twice as wide.
Step 2. $\sin(2\theta) = 0$ gives $2\theta = 0°, 180°, 360°, 540°$; halving each gives four angles. $\Rightarrow$ $\boldsymbol{\theta = 0°,\ 90°,\ 180°,\ 270°}$

Try it: For $r = 5\cos\theta$, solve $5\cos\theta = 5$ on $[0°, 360°)$.

check: $\boldsymbol{\theta = 0°}$

Skill 3 — Identify the family, then sketch it

Recognise the form before plotting:

- $r = a\cos\theta$ or $a\sin\theta$: a **circle** through the pole, offset along the 0° or 90° ray.
- $r = a\sin(k\theta)$ or $a\cos(k\theta)$: a **rose** — k petals when k is odd, $2k$ petals when k is even, each of length $|a|$.
- $r = a \pm b\cos\theta$: a **limacon**. Equal a and b gives a cardioid (cusp); $a/b < 1$ gives an inner loop; $a/b > 1$ gives no loop.
- Then sketch from four values: $\theta = 0°, 90°, 180°, 270°$, plus the zeros and the maximum.

Example: Sketch $r = 2 + 2\cos\theta$. Start with its greatest radius.
Step 1. Constant and coefficient are equal, so this is a cardioid; the sketch is anchored by the largest radius, which occurs where $\cos\theta$ is largest.
Step 2. $\cos\theta$ peaks at $\theta = 0°$, giving $r = 2 + 2(1)$; the curve then shrinks to 0 at 180°, the cusp. $\Rightarrow$ **4**

Try it: For $r = 3 + 3\sin\theta$, what is the greatest value of r?

check: **6**